Day 13 Study Notes

Wed 25 Nov 2026 · Sprint 60 · JEE Main 2027

Part	Time	What to do
Study	3 h 30 min	Physics: Ray optics (part 2: lenses) (75 min) · Maths: Binomial theorem (part 2) (75 min) · Chemistry: Hydrocarbons (part 1: alkanes and alkenes) (60 min)
Exam	1 h 15 min	The Day 13 exam in the Sprint 60 app: 30 questions, timed, no notes.
Review	1 h 15 min	Open every wrong and skipped question in the app, read its solution, tag the reason, then use Retry my mistakes.

How to use each chapter note. Read it once without stopping. Cover the formulas and write them from memory. Work the solved examples on paper before reading the working. Then read the NCERT section listed at the end.

Physics · Ray optics (part 2: lenses)

Lenses are the most asked part of ray optics. Today covers refraction at a curved surface, the lens formula, the lens maker's formula, power and combinations.

1. Refraction at a spherical surface

- $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$, with light going from medium n_1 into n_2, and the same sign convention as mirrors. R is positive if the centre of curvature is on the side the light goes to.

2. Thin lenses

- Lens formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$. Magnification: $m = \frac{v}{u}$. (Mirrors have $+$ in the formula and $-\frac{v}{u}$ for m; lenses the opposite.)
- Convex (converging) lens: f positive. Concave (diverging) lens: f negative.
- A concave lens always gives a virtual, erect, diminished image of a real object.
- A convex lens gives a real, inverted image when the object is beyond F, and a virtual, erect, magnified image when it is between F and the lens. At $2F$ the image is at $2F$, the same size.

3. Lens maker's formula

- $\frac{1}{f} = \left(\frac{n_{lens}}{n_{medium}} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$.
- Equiconvex lens with radius R and $n = 1.5$ in air: $f = R$.
- **Lens in a liquid:** the factor $\left(\frac{n_l}{n_m} - 1\right)$ shrinks, so f grows. If $n_m = n_l$ the lens disappears ($f = \infty$). If $n_m > n_l$ a convex lens starts to diverge.
- **Cutting a lens:** cut along the axis into two halves, each half keeps the focal length f (less light gets through). Cut perpendicular to the axis into two plano-convex lenses, each has $2f$.

4. Power and combinations

- Power $P = \frac{1}{f}$ with f in metres, measured in dioptres (D). Convex positive, concave negative.
- Thin lenses in contact: $P = P_1 + P_2$, that is $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$.
- Separated by distance d: $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$.
- For several lenses one after another, the total magnification is the product $m_1 m_2 \cdots$

5. Displacement method for f

- With object and screen a fixed distance D apart ($D \geq 4f$), a convex lens gives a sharp image at two positions a distance d apart.
- $f = \frac{D^2 - d^2}{4D}$. The two image sizes satisfy $h_o = \sqrt{h_1 h_2}$.
- The least distance between an object and its real image for a convex lens is $4f$.

Solved example 1. An object is 30 cm from a convex lens of focal length 20 cm. Find the image.
$u = -30$, $f = 20$: $\frac{1}{v} = \frac{1}{20} - \frac{1}{30} = \frac{1}{60}$, so $v = 60$ cm on the other side. $m = \frac{60}{-30} = -2$: real, inverted, twice the size.

Solved example 2. A glass lens ($n = 1.5$) has focal length 20 cm in air. Find its focal length in water ($n = \frac{4}{3}$).
$\frac{f_w}{f_a} = \frac{1.5 - 1}{\frac{1.5}{4/3} - 1} = \frac{0.5}{0.125} = 4$, so $f_w = 80$ cm.

Traps. Lens and mirror formulas differ in sign; do not mix them. Convert f to metres before finding power. A plano-convex lens has $R_2 = \infty$. The focal length of a lens changes in a liquid, but that of a mirror does not.

Mathematics · Binomial theorem (part 2)

Part 1 covered the general term. Today covers the sums of the binomial coefficients, the greatest coefficient, remainder and last-digit problems, and the number of terms in a multinomial expansion.

1. Sums of the coefficients

Write C_r for $\binom{n}{r}$, the coefficients of $(1+x)^n$.

- Put $x = 1$: $C_0 + C_1 + \cdots + C_n = 2^n$.
- Put $x = -1$: $C_0 - C_1 + C_2 - \cdots = 0$.
- So $C_0 + C_2 + C_4 + \cdots = C_1 + C_3 + C_5 + \cdots = 2^{n-1}$.
- $C_1 + 2C_2 + 3C_3 + \cdots + nC_n = n\,2^{n-1}$ (from $r\binom{n}{r} = n\binom{n-1}{r-1}$).
- $C_0^2 + C_1^2 + \cdots + C_n^2 = \binom{2n}{n}$.
- For any polynomial, the sum of all its coefficients is its value at $x = 1$. The sum of the coefficients of $(2x-1)^{20}$ is $1^{20} = 1$.

2. Greatest coefficient and greatest term

- The greatest binomial coefficient is the middle one: $\binom{n}{n/2}$ for even n; for odd n the two middle ones are equal and greatest.
- The numerically greatest **term** of $(1+x)^n$ depends on x. Compare $\dfrac{T_{r+1}}{T_r} = \dfrac{n-r+1}{r}\,|x|$ with 1: terms grow while this ratio exceeds 1.

3. Remainders and last digits

- Write the base as (multiple of the divisor) ± 1, or find a small power that is ± 1 modulo the divisor.
- Example: $2^{30} = (2^3)^{10} = 8^{10} = (7+1)^{10}$. Every term except the last has a factor 7, so the remainder on division by 7 is 1.
- Example: $5^2 = 25 = 26 - 1$, so $5^2 \equiv -1$ and $5^4 \equiv 1$ modulo 13.
- Last two digits: work modulo 100. $7^4 = 2401 \equiv 1$, so $7^{100} = (7^4)^{25}$ ends in 01.
- If the remainder comes out negative, add the divisor.

4. Multinomial expansions

- The number of terms in $(x_1 + x_2 + \cdots + x_k)^n$ is $\binom{n+k-1}{k-1}$. For $(a+b+c)^4$: $\binom{6}{2} = 15$.
- The coefficient of $a^p b^q c^r$ (with $p+q+r=n$) in $(a+b+c)^n$ is $\dfrac{n!}{p!\,q!\,r!}$.

5. Any index (for reference)

- For $|x| < 1$ and any real n: $(1+x)^n = 1 + nx + \dfrac{n(n-1)}{2!}x^2 + \ldots$ (an infinite series).
- $(1-x)^{-1} = 1 + x + x^2 + \ldots$ and $(1-x)^{-2} = 1 + 2x + 3x^2 + \ldots$

> **Solved example 1.** Find the remainder when 5^{99} is divided by 13.
> $5^4 \equiv 1 \pmod{13}$. $5^{99} = (5^4)^{24} \cdot 5^3 \equiv 125 \equiv 125 - 117 = 8$.

Solved example 2. Find $C_0^2 + C_1^2 + \cdots + C_4^2$ for $n = 4$.
$\binom{8}{4} = 70$. Check: $1 + 16 + 36 + 16 + 1 = 70$.

Traps. The sum of the coefficients means put $x = 1$, not count the terms. 2^n includes C_0. Remainders must lie between 0 and the divisor minus one. The greatest coefficient and the greatest term are different questions.

Chemistry · Hydrocarbons (part 1: alkanes and alkenes)

Hydrocarbons gives one question in most papers, usually on a named preparation, Markovnikov's rule, the peroxide effect or ozonolysis. Today covers alkanes and alkenes. Part 2 covers alkynes and benzene.

1. Alkanes: preparation

- **Wurtz reaction:** $2\mathrm{RX} + 2\mathrm{Na} \xrightarrow{\text{dry ether}} \mathrm{R{-}R} + 2\mathrm{NaX}$. Good for even-numbered symmetrical alkanes. Two different halides give a mixture of three alkanes (R–R, R–R′, R′–R′). Methane cannot be made this way.
- **Decarboxylation:** sodium salt of an acid heated with soda lime gives an alkane with one carbon fewer. $CH_3COONa \rightarrow CH_4$.
- **Kolbe electrolysis:** electrolysis of sodium or potassium salt of an acid gives R–R at the anode. Sodium acetate gives ethane.
- **Hydrogenation** of alkenes and alkynes with Ni, Pt or Pd.

2. Alkanes: reactions and conformations

- **Halogenation** goes by a free-radical chain (initiation, propagation, termination) in light. Reactivity $F_2 > Cl_2 > Br_2 > I_2$. Ease of replacing H: 3° > 2° > 1°.
- **Conformations of ethane:** staggered (more stable) and eclipsed. The energy difference is small (about 12.5 kJ/mol), so they interconvert rapidly and cannot be separated at room temperature.
- Boiling point rises with chain length and falls with branching.

3. Alkenes: preparation

- **Dehydrohalogenation:** alkyl halide with alcoholic KOH. **Saytzeff's rule:** the major product is the more substituted alkene. 2-Bromobutane gives mainly but-2-ene.
- **Dehydration of alcohols** with concentrated H_2SO_4; ease 3° > 2° > 1°.
- **Partial reduction of alkynes:** Lindlar's catalyst gives cis alkenes; Na in liquid NH_3 gives trans alkenes.
- Dehalogenation of vicinal dihalides with zinc.

4. Alkenes: reactions

- **Markovnikov's rule:** in addition of HX to an unsymmetrical alkene, H goes to the carbon that already has more hydrogens. The reaction goes through the more stable carbocation. Propene + HBr → 2-bromopropane.
- **Peroxide effect (Kharasch):** with HBr in the presence of a peroxide, addition goes the other way (anti-Markovnikov), by a free-radical route. Propene + HBr + peroxide → 1-bromopropane. Only HBr does this; HCl and HI do not.
- **Hydration** with dilute H_2SO_4 gives an alcohol following Markovnikov's rule.
- **Bromine water** is decolourised: a test for unsaturation.
- **Cold dilute alkaline $KMnO_4$** (Baeyer's reagent) gives a diol and loses its pink colour. Hot acidic $KMnO_4$ breaks the double bond into ketones, acids or CO_2.
- **Ozonolysis** (O_3, then Zn/H_2O) breaks C=C into two C=O groups. Replace the double bond with two C=O and read off the products. But-2-ene gives two molecules of ethanal; 2-methylpropene gives propanone and methanal.
- **Polymerisation:** ethene to polythene.

Solved example 1. What does 2-methylbut-2-ene give on ozonolysis?
$(CH_3)_2C{=}CHCH_3 \rightarrow (CH_3)_2C{=}O$ (propanone) + CH_3CHO (ethanal).

Solved example 2. Give the products of propene with HBr, with and without peroxide.
Without peroxide: $CH_3CHBrCH_3$ (Markovnikov). With peroxide: $CH_3CH_2CH_2Br$.

Traps. The peroxide effect works only with HBr. Wurtz with two different halides gives three alkanes, not one. Decarboxylation loses one carbon; Kolbe electrolysis doubles the alkyl group. Saytzeff applies to elimination, Markovnikov to addition.